In [12]:
```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
```

## Reading the CSV file

In [13]:
```
df = pd.read_csv("https://raw.githubusercontent.com/shrikant-temburw
df.head()
```

Out[13]:

| | Id | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | Species |
|---|---|---|---|---|---|---|
| 0 | 1 | 5.1 | 3.5 | 1.4 | 0.2 | Iris-setosa |
| 1 | 2 | 4.9 | 3.0 | 1.4 | 0.2 | Iris-setosa |
| 2 | 3 | 4.7 | 3.2 | 1.3 | 0.2 | Iris-setosa |
| 3 | 4 | 4.6 | 3.1 | 1.5 | 0.2 | Iris-setosa |
| 4 | 5 | 5.0 | 3.6 | 1.4 | 0.2 | Iris-setosa |

In [14]:
```
df.describe()
```

Out[14]:

| | Id | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm |
|---|---|---|---|---|---|
| count | 150.000000 | 150.000000 | 150.000000 | 150.000000 | 150.000000 |
| mean | 75.500000 | 5.843333 | 3.054000 | 3.758667 | 1.198667 |
| std | 43.445368 | 0.828066 | 0.433594 | 1.764420 | 0.763161 |
| min | 1.000000 | 4.300000 | 2.000000 | 1.000000 | 0.100000 |
| 25% | 38.250000 | 5.100000 | 2.800000 | 1.600000 | 0.300000 |
| 50% | 75.500000 | 5.800000 | 3.000000 | 4.350000 | 1.300000 |
| 75% | 112.750000 | 6.400000 | 3.300000 | 5.100000 | 1.800000 |
| max | 150.000000 | 7.900000 | 4.400000 | 6.900000 | 2.500000 |

In [15]:
```
df.shape
```

Out[15]:
```
(150, 6)
```

In [16]:
```
df["Species"].unique()
```

Out[16]:
```
array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=o
bject)
```

In [17]:
```
df.groupby("Species").size()
```

Out[17]:
```
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
dtype: int64
```

## List down the features and their types

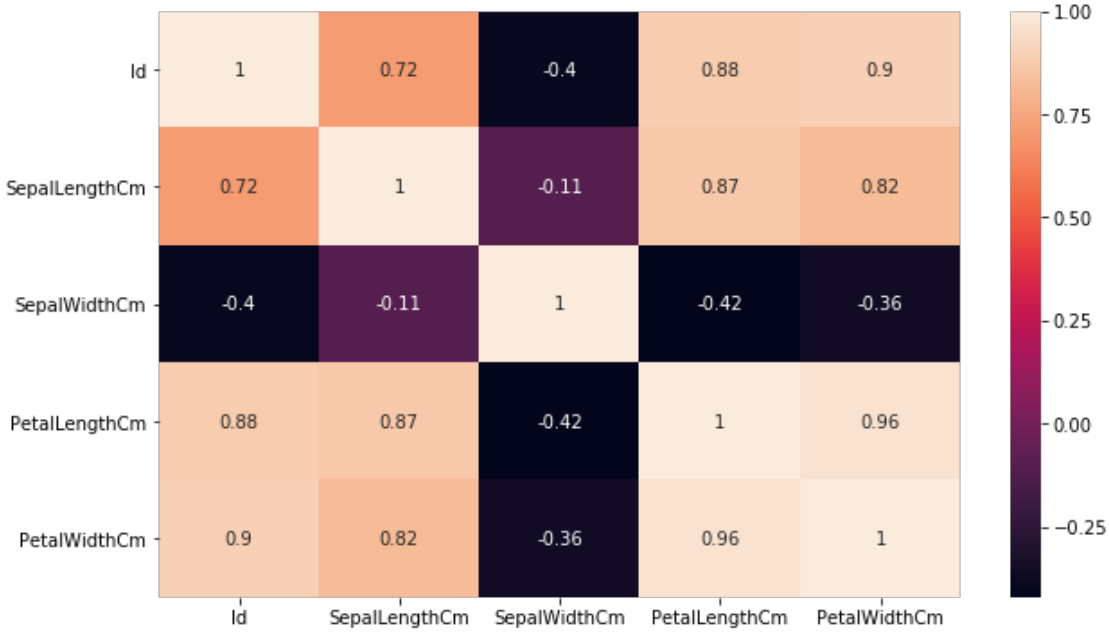
In [18]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
Id               150 non-null int64
SepalLengthCm    150 non-null float64
SepalWidthCm     150 non-null float64
PetalLengthCm    150 non-null float64
PetalWidthCm     150 non-null float64
Species          150 non-null object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.1+ KB
```

## Create a heatmap for each feature in the dataset.

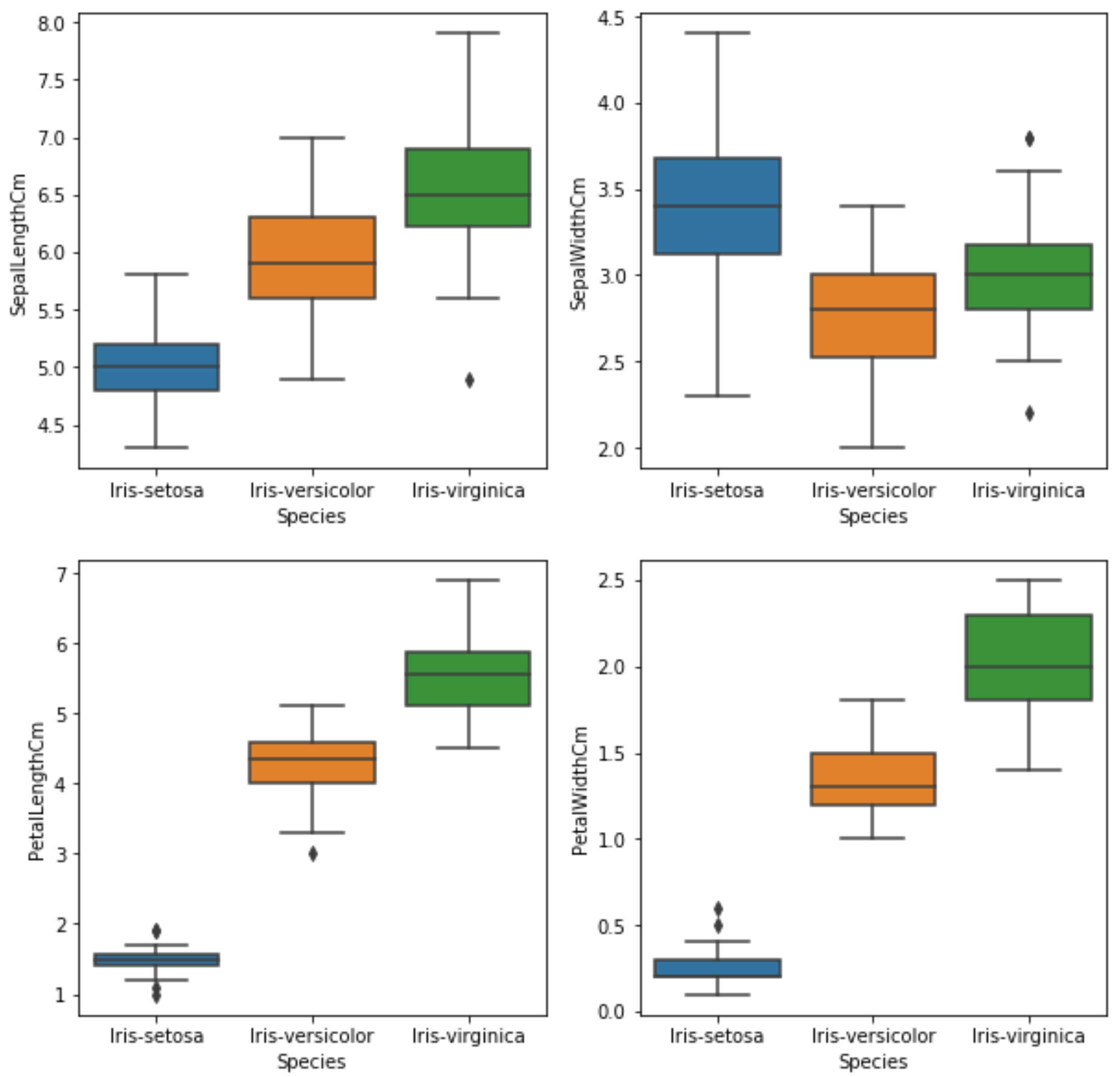
In [19]:
```
corr = df.corr()
plt.subplots(figsize=(10,6))
sns.heatmap(corr, annot=True)
```

Out[19]: <matplotlib.axes._subplots.AxesSubplot at 0x274aa714a90>



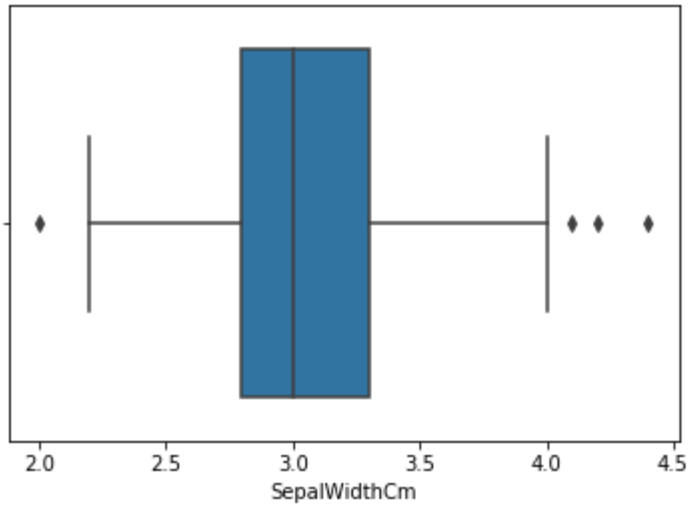

## Create a box plot for each feature in the dataset

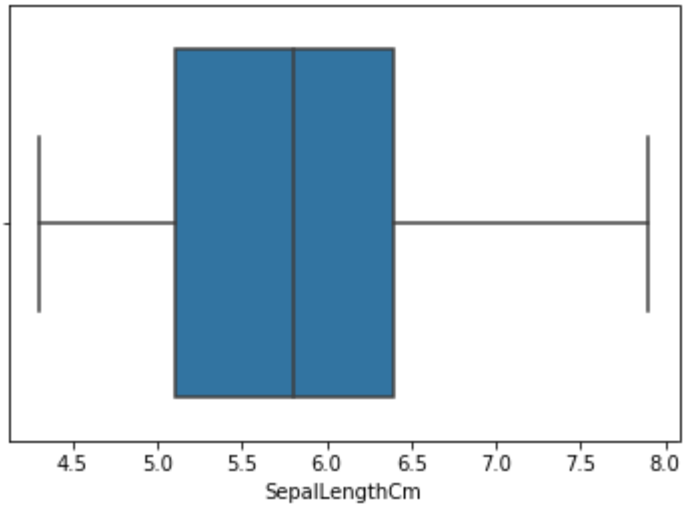
In [20]:
```
def graph(y):
 sns.boxplot(x="Species", y=y, data=df)

plt.figure(figsize=(10,10))

# Adding the subplot at the specified
# grid position
plt.subplot(221)
graph('SepalLengthCm')

plt.subplot(222)
graph('SepalWidthCm')

plt.subplot(223)
graph('PetalLengthCm')

plt.subplot(224)
graph('PetalWidthCm')

plt.show()
```

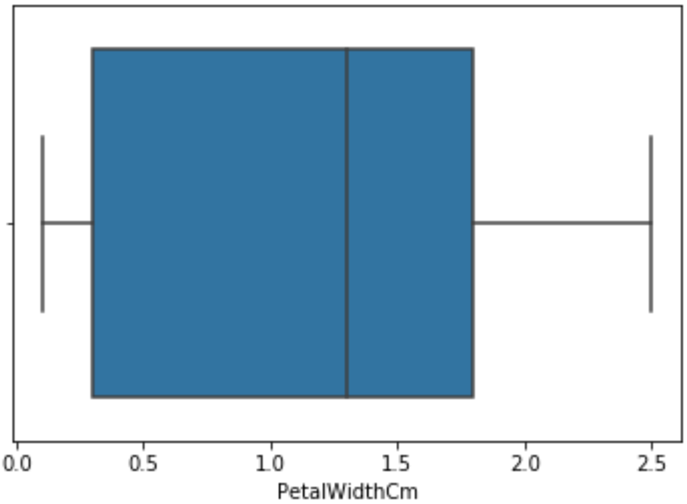


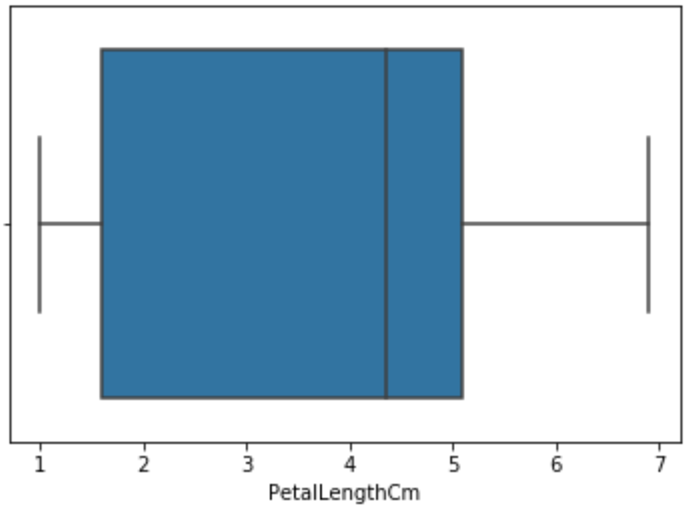

## Compare distributions and identify outliers.

In [21]:
```
sns.boxplot(x='SepalWidthCm', data=df)
plt.show()
sns.boxplot(x='SepalLengthCm', data=df)
plt.show()
sns.boxplot(x='PetalWidthCm', data=df)
plt.show()
sns.boxplot(x='PetalLengthCm', data=df)
plt.show()
```






In [ ]:

In [ ]: